Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2011-I-Q04] | Part A (Q1-Q5)

Answer: 056

1. Reflect C across the B-bisector to P on ray BA, giving $BP = BC = 120$. Reflect it across the A-bisector to Q on ray AB, giving $AQ = AC = 117$. Both points lie within AB.

2. The perpendicular feet M, N are the midpoints of CP, CQ. Hence $MN = PQ/2$ by the midpoint theorem.

3. Along the 125-unit base, $PQ = 120 + 117 - 125 = 112$, so $MN = 56$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2011-II-Q04] | Part A (Q1-Q5)

Answer: 051

1. Angle-bisector division gives $BD/DC = 20/11$. Balance masses $11, 20$ at B, C, giving mass 31 at D.

2. Assign mass 31 to A, so the A, D balance point is M. Grouping A, C first shows their balance point P lies on BM.

3. The lever ratio is $CP/PA = 31/20$, giving 51.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2022-I-Q03] | Part A (Q1-Q5)

Answer: 242

1. Let the A-bisector reach CD at U. Parallel-line angles imply $DU = AD = 333$. In isosceles triangle ADU, the D-bisector is a median, so P is the midpoint of AU. Similarly Q is the midpoint of BV, where $CV = 333$.

2. Choose $D = (0, 0), C = (650, 0), A = (75, h), B = (575, h)$. Then $U = (333, 0)$ and $V = (317, 0)$, yielding $P = (204, h/2)$ and $Q = (446, h/2)$.

3. The points have equal height, so $PQ = 446 - 204 = 242$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2001-II-Q07] | Part B (Q6-Q10)

Answer: 725

1. Use $P=(0,0), Q=(0,90), R=(120,0)$. The original inradius is $(90+120-150)/2=30$, so its center is $(30,30)$ and the two tangent lines are $x=60, y=60$.

2. Triangles at R, Q have similarity scales $1/2, 1/3$, hence inradii $15, 10$. Their centers are $(75,15)$ and $(10,70)$.

3. The squared separation is $65^2+55^2=7250$, giving $n=725$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2002-I-Q10] | Part B (Q6-Q10)

Answer: 148

1. $AB = 12, AC = 37$, so $BC = 35$ and $[ABC] = 210$. The bisector theorem gives $BD : DC = 12 : 37$, hence $[ADC] = 210 \cdot 37/49 = 1110/7$.

2. In AEF, the same bisector gives $EG : GF = 3 : 10$. Moreover $[AEF] = 210(3/12)(10/37) = 525/37$, so $[AGF] = (10/13)(525/37) = 5250/481$.

3. Subtract to get $[DCFG] = 1110/7 - 5250/481 \approx 147.6567$. The nearest integer is 148.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2007-II-Q09] | Part B (Q6-Q10)

Answer: 259

1. Using a rectangle coordinate system, $BE = DF = \sqrt{63^2 + 84^2} = 105$, $BF = DE = 364$, and $EF = \sqrt{63^2 + 280^2} = 287$. Both triangles have semiperimeter 378.

2. Tangent lengths give $EP = 378 - 364 = 14$ and $FQ = 14$. Therefore $PQ = 287 - 28 = 259$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2016-I-Q06] | Part B (Q6-Q10)

Answer: 013

1. The point order is C, I, L, D. Let half-angles at A, C be u, v. The exterior angle DIA equals $u+v$, while $DAI = DAB + BAI = v+u$ by the equal arcs AD, DB. Thus $DA = DI = 5$.

2. Triangles DAL, DCA are similar: their angle at D agrees and $DAL = DCA$ by the inscribed-angle theorem. Consequently $DA^2 = DL \cdot DC$, giving $DC = 25/3$.

3. Subtract DI along the bisector: $IC = DC - DI = 25/3 - 5 = 10/3$. Hence $p+q = 13$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2024-II-Q10] | Part B (Q6-Q10)

Answer: 468

1. Euler's center identity gives $OI^2 = R^2 - 2Rr$. In right triangle OIA, $IA^2 = R^2 - OI^2 = 2Rr = 156$. Since $IA = r/\sin(A/2)$, $\sin^2(A/2) = 3/13$.

2. It follows that $\sin A = 2\sqrt{30}/13$, $a = 2R\sin A = 4\sqrt{30}$, and $s - a = r\cot(A/2) = 2\sqrt{30}$, hence $s = 6\sqrt{30}$.

3. Equate $rs = bc\sin A/2$: $bc = 2 \cdot 6 \cdot 6\sqrt{30}/(2\sqrt{30}/13) = 468$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2026-II-Q08] | Part B (Q6-Q10)

Answer: 245

1. Put $a = AB = BC, b = AC$ and $u = AI/b = CI/b$. Heron plus the right triangle at the incircle contact point gives $u^2 = a/(2a + b)$, with $u > 0$. The perimeter ratio becomes $125(1 - 2u^2)(2u + 1) = 6$.

2. This factors as $(10u - 7)(50u^2 + 60u + 17) = 0$. The quadratic factor is positive for $u > 0$, so $u = 7/10$, yielding $a/b = 49/2$.

3. Integer sides require $a = 49t, b = 2t$ for positive integer t. Then $AI = 7t/5$ is integral precisely when $5 \mid t$. Taking $t = 5$ gives sides $245, 245, 10$ and $7, 7, 10$, so the minimum is 245.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2009-I-Q12] | Part C (Q11-Q15)

Answer: 011

1. Line AB is tangent at D, so this circle is the incircle of ABI. Let base tangent lengths be $u = AD, v = BD$, and the equal tangent lengths from I be w. Its inradius is $CD/2 = \sqrt{uv}/2$, using the right-triangle altitude identity.

2. The semiperimeter is $s = u + v + w$. Equate area rs to Heron $\sqrt{suvw}$: $s^2uv/4 = suvw$, so $s = 4w$. Thus $AB = u + v = 3w$.

3. The perimeter ratio is $2s/AB = 8/3$, giving 11.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2010-I-Q15] | Part C (Q11-Q15)

Answer: 045

1. Let $k = AM/CM$ and $d = BM$. The area ratio is k; equal area-to-semiperimeter ratios give $(12 + AM + d)/(13 + CM + d) = k$. Since $AM = kCM$, this reduces to $d = (13k - 12)/(1 - k) > 0$, requiring $12/13 < k < 1$.

2. Using $AM = 15k/(1 + k), CM = 15/(1 + k)$, Stewart gives $d^2 = (144 + 169k)/(1 + k) - 225k/(1 + k)^2$.

3. Equate the two formulas for d^2 and clear positive denominators. The result is $4k(69k^2 - 112k + 44) = 0$, with roots $0, 2/3, 22/23$. Only $22/23$ lies in the required interval.

4. It is attainable: $AM = 22/3, CM = 23/3, d = 10$ satisfy Stewart and the perimeter ratio. Hence $p + q = 45$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2012-I-Q12] | Part C (Q11-Q15)

Answer: 018

1. In triangle DCB, CE bisects its sixty-degree angle, so $CD/CB = DE/EB = 8/15$. Scale $CD = 8, CB = 15$.

2. The cosine rule gives $BD^2 = 64 + 225 - 120 = 169$, hence $BD = 13$. Applying it at B gives $\cos B = (169 + 225 - 64)/(2 \cdot 13 \cdot 15) = 11/13$.

3. Thus $\sin B = 4\sqrt{3}/13$, so $\tan B = 4\sqrt{3}/11$. The answer is 18.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2014-II-Q14] | Part C (Q11-Q15)

Answer: 077

1. Let the bisector meet the circumcircle again at E. It is the midpoint of arc BC not containing A, so its perpendicular projection onto line BC is M.

2. Projection takes collinear points A, P, E to H, N, M and preserves affine ratios. Since N is the midpoint, P is the midpoint of AE.

3. The angle at B is 105°. Arc angles give $\angle CBE = 15^\circ$, hence $\angle ABE = 120^\circ$, while $\angle AEB = 45^\circ$. The sine rule gives $AE = 10 \sin 120^\circ / \sin 45^\circ = 5\sqrt{6}$.

4. Thus $AP^2 = AE^2/4 = 75/2$, and the answer is 77.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2015-I-Q11] | Part C (Q11-Q15)

Answer: 108

1. Let $AB = t, BC = c$, and let M be the base midpoint. In the right triangles ABM, IBM, $\cos B = c/(2t)$ and $\cos(B/2) = c/16$. The double-angle identity gives $t = 64c/(c^2 - 128)$.

2. Positivity requires $c > \sqrt{128}$, while nondegeneracy and $BM < BI$ give $c < 16$. Thus only $c = 12, 13, 14, 15$ need testing.

3. The corresponding t values are $48, 832/41, 224/17, 960/97$. Only 48 is integral, and sides $48, 48, 12$ form a valid triangle with the derived half-angle relation. Its perimeter is 108.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2018-I-Q13] | Part C (Q11-Q15)

Answer: 126

1. The angle I_1AI_2 is constant at $A/2$. Put $\alpha = \angle AXB$. Incenter angles and the sine rule give $AI_1 = 30\sin(B/2)/\cos(\alpha/2)$ and $AI_2 = 34\sin(C/2)/\sin(\alpha/2)$.

2. Thus area equals $30 \cdot 34\sin(A/2)\sin(B/2)\sin(C/2)/\sin\alpha$. It is bounded below by its constant numerator. The altitude foot from A lies inside BC because its distance from B is $(30^2 + 32^2 - 34^2)/64 = 12$, so $\alpha = 90^\circ$ attains that bound.

3. For $a = 32, b = 34, c = 30, s = 48$, the half-angle product is $(s-a)(s-b)(s-c)/(abc)$. The minimum is therefore $16 \cdot 14 \cdot 18/32 = 126$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2020-I-Q13] | Part C (Q11-Q15)

Answer: 036

1. Let M be the midpoint of AD. In the circle through A, B, D, the bisector at B meets the midpoint of arc AD away from B; this is E, because it is equidistant from A, D. Thus $\angle EAD = B/2$. Similarly $\angle FAD = C/2$, on the opposite side.

2. Therefore $[AEF] = \frac{1}{2}AM^2(\tan(B/2) + \tan(C/2))$. The bisector theorem gives $BD = 2, DC = 3$, and $AD^2 = AB \cdot AC - BD \cdot DC = 18$, so $AM^2 = 9/2$.

3. Cosine law gives $\cos B = 1/8$, $\cos C = 3/4$. Hence the half-angle tangents are $\sqrt{7}/3$ and $1/\sqrt{7}$, with sum $10\sqrt{7}/21$.

4. The area is $15\sqrt{7}/14$, so $15 + 7 + 14 = 36$.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2022-I-Q14] | Part C (Q11-Q15)

Answer: 459

1. To prove the midpoint-cleaver lemma, put C at the origin and use unit vectors u along CA and v along CB. Write $a = CB, b = CA, c = AB$ and $M = (bu + av)/2$. If $a \geq b$, let $X = M - b(u+v)/2 = ((a-b)/2)v$, a point on CB. Then MX is parallel to $u+v$, the internal bisector direction at C. The boundary route $M - A - C - X$ has length $c/2 + b + (a - b)/2 = (a+b+c)/2$; the other route $M - B - X$ has the same length $c/2 + a - (a-b)/2$. Their common cut segment adds equally to both polygon perimeters. If $b > a$, swap the roles and choose $X = ((b-a)/2)u$. Thus the midpoint perimeter bisector is parallel to the opposite internal angle bisector.

2. Apply the lemma at the midpoints of AB and AC: the two lines are parallel to the internal bisectors at C and B. Their acute angle is $90^\circ - A/2$, so the given 30° forces $A = 120^\circ$.

3. Put $b = AC, c = AB, p = b + c, k = |b - c|$. The cosine law gives $219^2 = b^2 + c^2 + bc$, hence $k^2 = 4 \cdot 219^2 - 3p^2$. Modulo 3, then modulo 9, both k and p are multiples of 3.

4. Write $p = 3r, k = 3u$. Then $u^2 + 3r^2 = 146^2$. Triangle inequality and a nonnegative discriminant restrict r to $74, 75, \ldots, 84$. The values $146^2 - 3r^2$ are $4888, 4441, 3988, 3529, 3064, 2593, 2116, 1633, 1144, 649, 148$; only $2116 = 46^2$ is a square.

5. Thus $r = 80, p = 240, k = 138$, so the side lengths are $189, 51, 219$. They satisfy the strict triangle inequality and the cosine equation. The perimeter is 459.

Source / 来源

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2022-II-Q11] | Part C (Q11-Q15)

Answer: 180

1. Put $A=(0,0), D=(7,0)$ and the midpoint $M=(u,h)$ above the axis. Reflection across AM sends B to $(2,0)$; reflection across DM sends C to $(4,0)$. Since $MB=MC$, this implies $u=3$.

2. The half-angle tangents at A, D are $h/3, h/4$. Thus $B=(2(9-h^2)/(9+h^2), 12h/(9+h^2))$ and $C=(7-3(16-h^2)/(16+h^2), 24h/(16+h^2))$.

3. The condition $B_x+C_x=6$ reduces to $2h^2(h^2-5)=0$. Convexity gives $h>0$, hence $h=\sqrt{5}$. The coordinates become $B=(4/7, 6\sqrt{5}/7), C=(38/7, 8\sqrt{5}/7)$, and their midpoint is indeed $(3,\sqrt{5})$.

4. The shoelace formula gives area $6\sqrt{5}$; its square is 180. Both stated angles are acute for these coordinates.

Source / 来源